

# Spring AI: Building Intelligent Applications

From basic chat to advanced RAG and MCP →



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# What You'll Learn

- **Basic AI Integration:** ChatClient fundamentals
- **Streaming Responses:** Real-time AI interactions
- **Structured Data:** AI-powered object extraction
- **Multimodal AI:** Vision and audio capabilities
- **Function Calling:** Extend AI with custom tools
- **RAG Systems:** Knowledge-augmented AI
- **MCP Protocol:** Model Context Protocol implementation
- **Production Patterns:** Enterprise-ready architectures



# Repository Structure

```
1  Spring_AI_Training_Course/
2  |— labs.md           # 15 progressive lab exercises
3  |— src/
4  |   |— main/java/    # Service implementations
5  |   |— main/resources/ # Configuration & templates
6  |   |— test/java/    # Test-driven exercises
7  |— README.md         # Course documentation
8  |— slides.md         # This presentation
```

- **Start:** `main` branch with guided TODOs
- **Reference:** `solutions` branch when needed
- **Learn by doing:** Implement each lab incrementally
- **15 Labs:** From basic chat to advanced MCP servers

# Teaching Path

## Core

- Labs 1-6
- Lab 10
- Lab 12

## Demo

- Labs 7-9
- Multimodal APIs
- Audio/image costs

## Advanced

- Lab 13 Redis
- Labs 14-15 MCP
- Agent preview

# Spring AI Ecosystem

## AI Providers

- OpenAI (GPT, DALL-E)
- Anthropic (Claude)
- Azure OpenAI
- Google Vertex AI
- Local models (Ollama)

## Vector Stores

- SimpleVectorStore (in-memory)
- Redis Vector Store
- Pinecone, Weaviate
- PgVector, Chroma

## Capabilities

- Text generation
- Image analysis/generation
- Speech-to-text/text-to-speech
- Function calling
- RAG workflows

# Prerequisites

## Technical Requirements

- **Java 17+**
- **Spring Boot 3.5.14**
- **Spring AI 1.1.7**
- **Git** for branch management
- **Redis** (optional, for advanced RAG)

## Environment Setup

```
1  # Required API keys
2  export OPENAI_API_KEY=your_key
3  export ANTHROPIC_API_KEY=your_key
4
5  # Optional: Redis for advanced labs
6  docker run -p 6379:6379 redis/redis-stack:latest
7
8  # Clone and start
9  git clone <repo-url>
10 ./scripts/check-course-env
11 ./gradlew compileJava compileTestJava
```

# Lab 1-3: Foundations

Building Your First AI-Powered Spring Application

# Lab 1: Basic Chat Interactions

```
1 // Step 2: Complete implementation
2 @Service
3 public class ChatService {
4     private final ChatClient chatClient;
5
6     public ChatService(ChatClient.Builder builder) {
7         this.chatClient = builder.build();
8     }
9
10    public String generateResponse(String prompt) {
11        return chatClient.prompt(prompt)
12            .call()
13            .content();
14    }
15 }
```

**Result:** AI-powered responses in your Spring application! 🎉

## Lab 2: Request/Response Logging

```
1  @Service
2  public class ChatService {
3      private final ChatClient chatClient;
4
5      public ChatService(ChatClient.Builder builder) {
6
7          // Add logging advisor for debugging
8          this.chatClient = builder
9              .defaultAdvisors(new SimpleLoggerAdvisor())
10             .build();
11     }
12
13     public String generateResponse(String prompt) {
14         return chatClient.prompt(prompt)
15             .call()
16             .content();
17     }
18 }
```

**Debug Output:** See exactly what's sent to and received from AI models

# Lab 3: Streaming Responses

```
1  @RestController
2  public class ChatController {
3
4      @GetMapping(value = "/chat/stream", produces = MediaType.TEXT_EVENT_STREAM_VALUE)
5      public Flux<String> streamChat(@RequestParam String message) {
6          return chatClient.prompt(message)
7              .stream()
8              .content();
9      }
10
11      // Frontend receives real-time token-by-token responses
12      // Perfect for chat interfaces and long AI responses
13      // Uses Spring WebFlux Reactor streams
14  }
```

**Experience:** Real-time AI responses like ChatGPT interface

# Lab 4-6: Structured AI

From Text to Objects

# Lab 4: Structured Data Extraction

```
1  // Define your data structure
2  public record ActorFilms(String actor, List<String> movies) {}
3
4  @Test
5  void shouldGetActorFilms() {
6      // AI converts natural language to structured data
7      ActorFilms actorFilms = chatClient.prompt("Generate the filmography for Tom Hanks")
8          .call()
9          .entity(ActorFilms.class);
10
11     // Assert AI returned proper structure
12     assertThat(actorFilms.actor()).isEqualTo("Tom Hanks");
13     assertThat(actorFilms.movies()).contains("Forrest Gump", "Cast Away");
14 }
```

**Magic:** AI understands your Java objects and populates them correctly!

# Lab 5: Prompt Templates

```
1  @Component
2  public class TemplateService {
3
4      @Value("classpath:/prompts/actor-filmography.st")
5      private Resource actorFilmographyTemplate;
6
7      public ActorFilms getActorFilms(String actorName) {
8          return chatClient.prompt()
9              .user(userSpec -> userSpec
10                  .text(actorFilmographyTemplate)
11                  .param("actor", actorName)
12                  .param("count", 5))
13              .call()
14              .entity(ActorFilms.class);
15      }
16  }
```

## Template File ( actor-filmography.st ):

```
1  Generate a filmography for {actor}.
2  Include exactly {count} of their most famous movies.
3  Format as JSON with actor name and movies array.
```

# Lab 6: Chat Memory

```
1  @Service
2  public class ConversationService {
3      private static final String CONVERSATION_ID = "course-demo";
4      private final ChatClient chatClient;
5
6      public ConversationService(ChatClient.Builder builder, ChatMemory memory) {
7          this.chatClient = builder
8              .defaultAdvisors(MessageChatMemoryAdvisor.builder(
9                  memory).build())
10             .build();
11     }
12
13     public String continueConversation(String message) {
14         return chatClient.prompt()
15             .advisors(a -> a.param(
16                 ChatMemory.CONVERSATION_ID, CONVERSATION_ID))
17             .user(message)
18             .call()
19             .content();
20         // AI remembers previous messages in this conversation!
21     }
22 }
```

**Try this:**

# Lab 7-9: Multimodal AI

## Beyond Text: Vision and Audio

```
RUN_MULTIMODAL_TESTS=true ./gradlew test --tests OpenAiTests --tests AudioTests
```

# Lab 7: Vision Capabilities

```
1  @Test
2  void shouldAnalyzeImage() {
3      var imageResource = new ClassPathResource(
4          "/images/multimodal_test_image.png");
5
6      String response = chatClient.prompt()
7          .user(userSpec -> userSpec
8              .text("What do you see in this image?")
9              .media(MimeTypeUtils.IMAGE_PNG, imageResource))
10         .call()
11         .content();
12
13      assertThat(response.toLowerCase())
14          .contains("dog", "playing");
15  }
```

# Vision: What AI Can Analyze

- **Photos and diagrams**
- **Charts and graphs**
- **Screenshots and UI mockups**
- **Medical images**
- **Technical drawings**



# Lab 8: Image Generation

```
1  @Service
2  public class ImageService {
3      private final ImageModel imageModel;
4
5      public ImageService(ImageModel imageModel) {
6          this.imageModel = imageModel;
7      }
8
9      public String generateImage(String prompt) {
10         ImageResponse response = imageModel.call(
11             new ImagePrompt(prompt,
12                 OpenAiImageOptions.builder()
13                     .model("gpt-image-1")
14                     .build()));
15
16         // gpt-image-1 returns base64-encoded images
17         return response.getResult()
18             .getOutput()
19             .getB64Json();
20     }
21 }
```

**Create:** AI-generated images from text descriptions

# Lab 9: Audio Capabilities

```
1  @Service
2  public class SpeechService {
3      private final OpenAiAudioSpeechModel speechModel;
4
5      public SpeechService(OpenAiAudioSpeechModel speechModel) {
6          this.speechModel = speechModel;
7      }
8
9      public byte[] generateSpeech(String text) {
10         var options = OpenAiAudioSpeechOptions.builder()
11             .voice(OpenAiAudioApi.SpeechRequest.Voice.ALLOY)
12             .responseFormat(OpenAiAudioApi.SpeechRequest
13                 .AudioResponseFormat.MP3)
14             .speed(1.0)
15             .build();
16
17         var prompt = new TextToSpeechPrompt(text, options);
18         return speechModel.call(prompt).getResult().getOutput();
19     }
20 }
```

**Output:** High-quality AI-generated speech from text

# Lab 9: Speech-to-Text

```
1  @Service
2  public class AudioService {
3      private final AudioTranscriptionModel transcriptionModel;
4
5      public AudioService(AudioTranscriptionModel model) {
6          this.transcriptionModel = model;
7      }
8
9      public String transcribeAudio(Resource audioFile) {
10         var prompt = new AudioTranscriptionPrompt(audioFile);
11         var response = transcriptionModel.call(prompt);
12         return response.getResult().getOutput();
13     }
14 }
```

**Capability:** Convert speech files (MP3, WAV) to accurate text transcription

# Lab 10-11: Application Patterns

Tools and Production Refactoring

# Lab 10: AI Tools (Function Calling)

```
1 // Step 3: AI automatically calls your tools
2 @Test
3 void shouldCallTool() {
4     String response = chatClient.prompt()
5         .user("What time is it right now?")
6         .tools(new DateTimeTools())
7         .call()
8         .content();
9
10    // AI called getCurrentDateTime() automatically!
11    assertThat(response).contains("2024");
12 }
```

**Result:** AI can execute your Java methods when needed!

layout: section

# Lab 12-13: RAG Systems

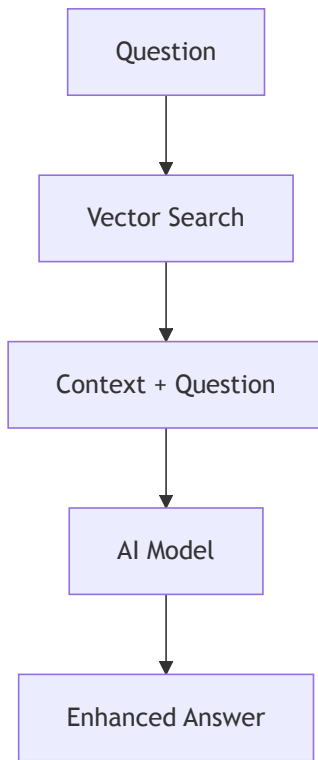
## Knowledge-Augmented AI

# Lab 12: The RAG Problem

## Traditional AI Limitations

- Knowledge cutoff dates
- Can't access your documents
- No real-time information
- Generic responses only

## RAG Solution



# RAG: How It Works

1. **Document Processing:** Split documents into chunks
2. **Embedding Generation:** Convert chunks to vectors
3. **Vector Storage:** Store in searchable database
4. **Query Processing:** Find relevant chunks for question
5. **Context Enhancement:** Add found content to AI prompt
6. **Enhanced Response:** AI answers using your data

**Result:** AI answers using YOUR documents and data! 🎯

# Spring AI: Supported Providers

## Chat Models

- OpenAI • Azure OpenAI
- Anthropic • Google VertexAI
- Amazon Bedrock • Ollama
- Hugging Face • Mistral AI
- Groq • NVIDIA • Perplexity
- DeepSeek • Moonshot AI
- QianFan • ZhiPu AI • MiniMax

## Embedding Models

- OpenAI • Azure OpenAI
- Amazon Bedrock • VertexAI
- Ollama • Mistral AI
- PostgresML • ONNX
- QianFan • ZhiPu AI
- OCI GenAI • MiniMax

## Image & Audio

- **Images:** OpenAI DALL-E
- **Images:** Stability AI, ZhiPu AI
- **Speech-to-Text:** OpenAI  
Whisper
- **Text-to-Speech:** OpenAI TTS
- **Moderation:** OpenAI, Mistral

**Spring AI advantage:** Portable API - switch providers with configuration only!

# Understanding Embeddings & Vector Search

## What are Embeddings?

- **Numerical representation** of text meaning
- **High-dimensional vectors** (typically 1536+ dimensions)
- **Semantic similarity** via distance calculations
- **Context-aware** - same words, different meanings

## Chunking Strategies

- **Token-based**: Split by token count (GPT tokenizer)
- **Sentence-based**: Preserve sentence boundaries
- **Semantic**: Split by topic/meaning changes
- **Overlapping**: Chunks share context at boundaries

# RAG Implementation

```
1 // Step 3: Testing RAG
2 @Test
3 void shouldAnswerFromDocuments() {
4     String answer = ragService.askQuestion(
5         "What is Spring AI ChatClient?");
6
7     // AI uses loaded PDF content to answer!
8     assertThat(answer).contains("ChatClient", "Spring AI");
9 }
```

**Magic:** AI answers questions using your PDF documents!

# Lab 13: Production RAG with Redis

```
1  @Configuration
2  @Profile("redis")
3  public class RedisRAGConfig {
4
5      @Bean
6      public VectorStore vectorStore(RedisConnectionFactory factory) {
7          return new RedisVectorStore(factory, embeddingModel());
8      }
9
10     @Bean
11     public ApplicationRunner dataLoader(VectorStore vectorStore) {
12         return args -> loadDocumentsIfEmpty(vectorStore);
13     }
14 }
```

# Redis RAG: Smart Data Loading

```
1  @Bean
2  public ApplicationRunner dataLoader(VectorStore vectorStore) {
3      return args -> {
4          if (vectorStore instanceof RedisVectorStore redis &&
5              redis.getCollection().isEmpty()) {
6
7              // Only load documents if Redis is empty
8              loadDocuments(vectorStore);
9              log.info("Loaded {} documents into Redis", count);
10         }
11     };
12 }
```

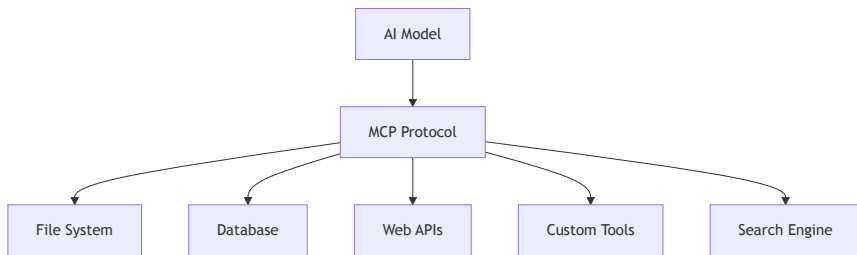
**Benefits:** Persistent • Scalable • Fast similarity search

# Lab 14-15: Model Context Protocol

The Future of AI Tool Integration

# What is MCP?

- **Standardized protocol** for AI tool communication
- **Open source** by Anthropic
- **Universal interface** between AI and tools
- **Secure sandbox** for AI operations
- **Growing ecosystem** of MCP servers



MCP enables AI to securely access external tools and data sources

# Lab 14: MCP Client

```
1  @SpringBootTest
2  @ActiveProfiles("mcp")
3  @EnabledIfEnvironmentVariable(
4      named = "RUN_MCP_CLIENT_TESTS",
5      matches = "true")
6  public class McpClientTests {
7      @Autowired
8      private ChatModel chatModel;
9
10     @Autowired(required = false)
11     private ToolCallbackProvider toolCallbackProvider; // MCP tool provider
12
13     private ChatClient chatClient;
14     private ToolCallback[] mcpTools;
15
16     @BeforeEach
17     void setUp() {
18         // Extract tools from the provider
19         if (toolCallbackProvider != null) {
20             mcpTools = toolCallbackProvider.getToolCallbacks();
21         }
22         // Build ChatClient with discovered tools
23         chatClient = ChatClient.builder(chatModel)
24             .defaultToolCallbacks(mcpTools)
```

# MCP Client: Configuration

**Configuration** ( application-mcp.properties ):

```
1  # Enable MCP client
2  spring.ai.mcp.client.enabled=true
3  spring.ai.mcp.client.toolcallback.enabled=true
4
5  # Context7 - Library documentation lookup
6  spring.ai.mcp.client.stdio.connections.context7.command=npx
7  spring.ai.mcp.client.stdio.connections.context7.args=-y,@upstash/context7-mcp@latest
8
9  # Tavily - AI-optimized web search (requires TAVILY_API_KEY)
10 spring.ai.mcp.client.stdio.connections.tavily.command=npx
11 spring.ai.mcp.client.stdio.connections.tavily.args=-y,tavily-mcp@latest
```

**Result:** 6 tools discovered - 2 from Context7, 4 from Tavily!

# Lab 15: MCP Server

```
1  # Step 3: Run MCP server
2  ./gradlew bootRun --args='--spring.profiles.active=mcp-server'
3
4  # Step 4: Connect from Claude Desktop
5  # Add to Claude's MCP config:
6  {
7    "mcpServers": {
8      "spring-calculator": {
9        "command": "java",
10       "args": ["-jar", "app.jar", "--spring.profiles.active=mcp-server"]
11     }
12   }
13 }
```

**Result:** Claude Desktop can use your Java methods as tools!

# Production Patterns

Enterprise-Ready Spring AI

# Service Layer: REST Controller

```
1  @RestController
2  @RequestMapping("/api/ai")
3  public class AIController {
4      private final ChatService chatService;
5
6      @PostMapping("/chat")
7      public ResponseEntity<String> chat(@RequestBody ChatRequest request) {
8          try {
9              String response = chatService.generateResponse(request.getMessage());
10             return ResponseEntity.ok(response);
11         } catch (Exception e) {
12             return ResponseEntity.status(500)
13                 .body("AI service temporarily unavailable");
14         }
15     }
}
```

# Service Layer: Streaming Support

```
1  @GetMapping("/chat/stream")
2  public ResponseEntity<Flux<String>> streamChat(@RequestParam String message) {
3      Flux<String> stream = chatService.streamResponse(message)
4          .onErrorReturn("Error occurred during streaming");
5
6      return ResponseEntity.ok()
7          .contentType(MediaType.TEXT_EVENT_STREAM)
8          .body(stream);
9  }
10
11  // Clean separation of concerns
12  // Proper error handling
13  // Stream-ready architecture
14  // Production-ready patterns
```

# Configuration Management

```
1  @Configuration
2  public class AIConfiguration {
3
4      @Bean
5      @Primary
6      @ConditionalOnProperty("spring.ai.openai.api-key")
7      public ChatModel primaryChatModel(OpenAiChatModel openAiModel) {
8          return openAiModel;
9      }
10
11     @Bean
12     @Profile("rag")
13     public VectorStore vectorStore(EmbeddingModel embeddingModel) {
14         return SimpleVectorStore.builder(embeddingModel).build();
15     }
16 }
```

# Profile-Based Feature Activation

```
1  @Bean
2  @Profile({"rag", "redis"})
3  public VectorStore redisVectorStore(RedisConnectionFactory factory) {
4      return new RedisVectorStore(factory, embeddingModel());
5  }
6
7  @Bean
8  @Profile("mcp")
9  public McpClientConfiguration mcpConfig() {
10     return new McpClientConfiguration();
11 }
```

## Run configurations:

```
1  ./gradlew bootRun                                # Basic AI
2  ./gradlew bootRun --args='--spring.profiles.active=rag,redis' # RAG
3  ./gradlew bootRun --args='--spring.profiles.active=mcp'      # MCP
```

# Error Handling: Retries

```
1  @Service
2  public class ResilientChatService {
3      private final ChatClient chatClient;
4
5      @Retryable(value = {ApiException.class}, maxAttempts = 3)
6      public String generateResponse(String prompt) {
7          return chatClient.prompt(prompt).call().content();
8      }
9
10     // Automatically retries API failures
11     // Exponential backoff available
12     // Works with Spring Retry
13 }
```

# Error Handling: Circuit Breaker

```
1  @CircuitBreaker(name = "ai-service", fallbackMethod = "fallbackResponse")
2  public String generateResponseWithCircuitBreaker(String prompt) {
3      return chatClient.prompt(prompt).call().content();
4  }
5
6  public String fallbackResponse(String prompt, Exception ex) {
7      log.warn("AI service failed, using fallback", ex);
8      return "I'm temporarily unable to process your request.";
9  }
10
11  // Prevents cascade failures
12  // Automatic recovery when service improves
```

# Error Handling: Async Processing

```
1  @Async
2  public CompletableFuture<String> generateResponseAsync(String prompt) {
3      return CompletableFuture.supplyAsync(() ->
4          chatClient.prompt(prompt).call().content());
5  }
6
7  // Usage
8  CompletableFuture<String> future = service.generateResponseAsync("Hello");
9  String result = future.get(30, TimeUnit.SECONDS);
```

**Benefits:** Non-blocking • Timeout control • Better UX

# Testing: Unit Tests with Mocks

```
1  @SpringBootTest
2  class ChatServiceTest {
3      @MockitoBean ChatClient chatClient;
4      @MockitoBean ChatClient.CallResponseSpec callSpec;
5
6      @Test
7      void shouldGenerateResponse() {
8          when(chatClient.prompt("Hello")).thenReturn(callSpec);
9          when(callSpec.call()).thenReturn(mockResponse("Hi there!"));
10
11         String result = service.generateResponse("Hello");
12         assertThat(result).isEqualTo("Hi there!");
13     }
14 }
```

# Testing: Integration with TestContainers

```
1  @Testcontainers
2  @SpringBootTest
3  class RAGIntegrationTest {
4
5      @Container
6      static RedisContainer redis = new RedisContainer("redis:7.0")
7          .withExposedPorts(6379);
8
9      @Test
10     void shouldPerformRAGWithRedis() {
11         // Test actual RAG workflow with real Redis
12         String answer = ragService.askQuestion("What is Spring AI?");
13         assertThat(answer).contains("Spring", "AI");
14     }
15 }
```

**Benefits:** Real dependencies • Isolated environments • CI/CD ready

# Cost & Performance Optimization

## Model Selection

- **Small models:** Fast, cheap, great for routing
- **Reasoning models:** Slower, better for complex tasks
- **Multimodal models:** Images, audio, documents
- **Local models:** Privacy, offline-friendly

## Optimization Strategies

- **Token Management:** Monitor usage, optimize prompts
- **Embedding Caching:** Store frequently used vectors
- **Request Batching:** Combine operations when possible
- **Smart Chunking:** Optimize document splitting

# Security & Observability

## Security Best Practices

- **API Key Management:** Environment variables, vaults
- **Data Privacy:** Local processing when possible
- **Input Validation:** Sanitize user prompts
- **Output Filtering:** Check AI responses

## Monitoring & Observability

- **Actuator + Micrometer:** Spring AI observations
- **Metrics:** Latency, token usage, vector store calls
- **Tracing:** ChatClient, advisors, model calls
- **Prompt logging:** Useful, but off by default

# Course Summary

What You've Built

# Your AI-Powered Application Stack

## Foundation & Advanced

- ChatClient integration
- Multiple AI providers
- Streaming responses
- Multimodal capabilities
- Function calling

## Production Ready

- RAG systems
- MCP protocol
- Service architecture
- Error handling
- Comprehensive testing

**Result:** Enterprise-grade Spring AI applications! 🚀

# Key Takeaways

1. **Spring AI simplifies AI integration** - No complex API calls or JSON parsing
2. **Start simple, add complexity gradually** - From basic chat to advanced RAG
3. **Leverage Spring's strengths** - Auto-configuration, profiles, testing
4. **Think beyond text** - Vision, audio, and structured data open new possibilities
5. **MCP is the future** - Standardized tool integration across AI platforms
6. **Production requires planning** - Error handling, monitoring, and resilience
7. **Test everything** - AI responses are non-deterministic but testable

# Next Steps: Continue Learning

## Hands-On Practice

- Explore the GitHub repository
- Try advanced RAG techniques
- Build custom MCP servers
- Integrate with existing apps

## Key Resources

- Spring AI Docs
- Model Context Protocol
- Course Repository

# Production Considerations

## Operational Excellence

- **API Cost Management:** Monitor token usage
- **Rate Limiting:** Handle API quotas
- **Data Privacy:** Keep sensitive data secure
- **Monitoring:** Track performance and errors

## Advanced Topics

- Custom embedding models
- AI agents (see next slides)
- AI-powered workflows
- Integration with existing systems

# AI Agents: The Landscape

## Spring AI Foundation

- **Tool calling** ( `@Tool` ) — you already learned this!
- **Recursive Advisors** — plan-execute-iterate loops
- **Effective Agents Guide** — patterns, not a framework
- Spring AI provides *building blocks*, not a prescriptive agent framework

## Agent Frameworks & Tools

- **Embabel** — Rod Johnson's agent framework *on top of* Spring AI
  - `@Agent` , `@Goal` , `@Action` annotations
  - Goal-Oriented Action Planning (from game AI)
- **spring-ai-agent-utils** — Claude Code-style tools for Spring AI (requires 2.0)
- **Spring AI Agents/Bench** — benchmarking & evaluation

# Embabel: Agent Pattern in Action

The framework infers execution order from input/output types — like a type-driven workflow

```
1 // The framework automatically plans: UserInput → Story → ReviewedStory
2 // No explicit orchestration needed – types drive the workflow
3 //
4 // Compare to @Tool (Lab 10): Tools extend what an LLM can do
5 // Agents orchestrate multiple LLM calls toward a goal
```

**Demo:** [github.com/kousen/OperaGenerator](https://github.com/kousen/OperaGenerator) ( `embabel` branch)

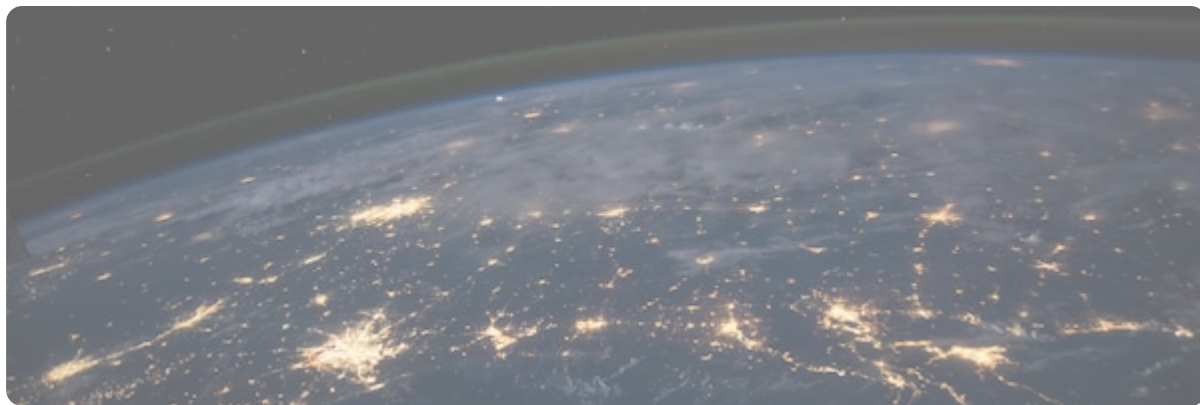
# Thank You!

## Questions?

**Kenneth Kousen**

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Ready to build intelligent applications with Spring AI!